UNITED INTERNATIONAL UNIVERSITY (UIU)

Dept. of Computer Science & Engineering

Class Test 1 (Set E)

Course No: CSE 4325 Course Title: Microprocessors and Microcontrollers

Time: 15 minutes **Marks: 20**

Name		**Section**		**ID**	

1. Transfer of bus control from the processor to the device takes **450 ns**, while transfer of bus control from the device to the processor takes **350 ns**. If one of the I/O devices uses DMA in **cycle stealing mode** and has a data transfer rate of **35 KB/s**, how long will it take to transfer a block of **540 bytes**? Assume that **6 bytes** are transferred at a time in this mode. (10)
2. Transfer of bus control from the processor to the device takes **600 ns**, while transfer of bus control from the device to the processor takes **500 ns**. One of the I/O devices uses DMA **in both burst mode and cycle stealing mode** and has a data transfer rate of **50 KB/s** For the **first half** of the bytes, the device uses burst mode and for the rest half, it uses cycle stealing mode. Assume that in cycle stealing mode, data is transferred **16 bytes** at a time. How many bytes are transferred in **80141900 ns**? (10)